

Dang Viet Anh Nguyen

PhD Candidate | Model-Based RL, World Models & Multi-Agent Control
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Research Profile

PhD researcher developing model-based learning and decision-making methods for large-scale multi-agent systems under uncertainty. My work combines graph neural world models, model-based reinforcement learning, model predictive control, and mathematical optimization for robust and adaptive planning. I develop action-conditioned world models, search-guided policy learning, and distributed control methods that connect learned dynamics with real-time decision-making. I am particularly interested in world models, learning-control integration, distribution shift, and multi-agent autonomy, with the goal of extending these methods toward physical autonomous systems.

Research Experience

Visiting Researcher

Technical University of Munich, Germany
March–September 2026

Host: Professor Constantinos Antoniou

Conduct research on model-based learning, planning, and control for large-scale networked systems, with traffic networks as the primary experimental domain. Develop and evaluate graph-based world models and model-based decision-making algorithms for adaptive multi-agent control under uncertainty, non-stationarity, and disruptions.

PhD Researcher in Machine Learning and Control

Technical University of Denmark, Denmark
2023–Present

Develop model-based learning, planning, and control methods for large-scale multi-agent systems operating under uncertainty and disruptions. Current work includes GNN-MPC, integrating hierarchical action-conditioned graph world models with distributed receding-horizon control and search-based coordination; GNN-AlphaZero, a search-guided approximate policy-iteration framework that uses world-model rollouts to train a scalable factorized policy-value network; and T-REX, a robustness benchmark for learning-based control under incidents and distribution shift. Conduct end-to-end research from algorithm design and large-scale experimentation to robustness analysis and MSc co-supervision.

Research Engineer — Intelligent Systems and Optimization

Singapore Management University, Singapore
2022–2023

Developed a hybrid reinforcement learning and combinatorial optimization framework for sequential decision-making in stochastic e-waste collection under uncertain requests and operational constraints, achieving a 24% improvement over state-of-the-art methods on real data. Led the full research pipeline from problem formulation and algorithm design to experimentation and operational integration, resulting in publications in EJOR, EvoCOP, ICCL, and IEEE CASE.

Education

2023 – Present (Expected graduate in Dec 2026)	PhD Candidate in Machine Learning & Control Technical University of Denmark (DTU), Denmark Project: Proactive traffic control through AI and Big Data Supervisors: Filipe Rodrigues , Carlos Lima Azevedo
2021-2022	MSc. Supply Chain and Logistics Nanyang Technological University (NTU), Singapore Thesis: Two-echelon Vehicle Routing Problem for Post-disaster Aid Advisor: Rajesh Piplani
2015-2020	BE in Transportation Engineering Hanoi University of Civil Engineering (HUCE), Vietnam Awards: Third Prize, Vietnam Olympiad on Mechanics; Student with Five Good Merits (National Level)

Research Interests

World Models & Model-Based Reinforcement Learning

Action-conditioned dynamics learning for predictive rollouts, policy improvement, and long-horizon decision-making.

Learning-Augmented Planning & Control

Integration of learned dynamics, model predictive control, mathematical optimization, and search.

Robust & Adaptive Autonomy

Generalization under distribution shift, disruptions, model uncertainty, and changing operating regimes.

Multi-Agent & Graph-Based Autonomous Systems

Scalable coordination, decentralized execution, cooperative control, and structured graph representations for networked dynamical systems.

Working Papers

Nguyen, D. V. A., Azevedo, C. L., Toledo, T., & Rodrigues, F. (2026). GNN-AlphaZero: Search-Guided Policy Iteration for Cooperative Traffic Signal Control. (Working paper)

Nguyen, D. V. A., Azevedo, C. L., Toledo, T., & Rodrigues, F. (2026). Disruption-Aware Learning of Traffic Network Dynamics for Model Predictive Signal Control. (Working paper)

Peer-reviewed Journal Articles

Nguyen, D. V. A., Azevedo, C. L., Toledo, T., & Rodrigues, F. (2025). A Framework for Benchmarking Traffic Signal Control Robustness under Incidents: Comparative Study of Reinforcement Learning-Based Methods. *European Transport Research Review* (Accepted).

Nguyen, D. V. A., Gunawan, A., Misir, M., Hui, L. K., & Vansteenwegen, P. (2025). Deep reinforcement learning for solving the stochastic e-waste collection problem. *European Journal of Operational Research*.

Nguyen, D. V. A., Flensburg, J. V., Cerreto, F., Pascariu, B., Pellegrini, P., Azevedo, C. L., & Rodrigues, F. (2026). Multi-Graph Inductive Representation Learning for large-scale Urban Rail demand prediction under disruptions. *Computers & Industrial Engineering*, 111924.

Peer-reviewed Conference Papers

Nguyen, D. V. A., Gunawan, A., Misir, M., & Vansteenwegen, P. (2024, April). Q-learning based framework for solving the stochastic e-waste collection problem. In *European Conference on Evolutionary Computation in Combinatorial Optimization (Part of EvoStar)* (pp. 49-64). Cham: Springer Nature Switzerland.

Gunawan, A., Hoe, S. L., Lim, X. Y., Tran, L. C., & **Nguyen, D. V. A.** (2023, December). ExploreLah: Personalised and Smart Trip Planner for Mobile Tourism. In *2023 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)* (pp. 0923-0927). IEEE.

Gunawan, A., **Nguyen, D. V. A.**, Nguyen, P. K. M., & Vansteenwegen, P. (2023, September). Grasp solution approach for the e-waste collection problem. In *International Conference on Computational Logistics* (pp. 260-275). Cham: Springer Nature Switzerland.

Gunawan, A., Nguyen, M. P. K., Vincent, F. Y., & **Nguyen, D. V. A.** (2023, August). The heterogeneous vehicle routing problem with multiple time windows for the e-waste collection problem. In *2023 IEEE 19th International Conference on Automation Science and Engineering (CASE)* (pp. 1-6). IEEE.

Manuscripts Under Review and Preprints

Nguyen, D. V. A., Alma F., Kiristine C., Toledo, T., & Rodrigues, F. (2025). Hierarchical Forecast Reconciliation for Urban Rail Transit Demand Prediction under Operational Disruptions. (Under review)

Nguyen, D. V. A., Piplani, R. & Gunawan, A. (2024). Two-echelon covering tour vehicle routing problem for post-disaster aid (Under review)

Nguyen, V. P., **Nguyen, D. V. A.** & Truong, L. T. (2024). Adoption Intentions across e-Mobility Modes: Electric Cars, Hybrid Electric Cars, Electric Motorbikes, and Electric Bikes. (Under review)

Presentations

Nguyen, D. V. A., Flensburg, J. V., Cerreto, F., Pascariu, B., Pellegrini, P., Azevedo, C. L., & Rodrigues, F. (2025) Large-scale demand prediction in urban rail using multi-graph inductive representation learning. *104th Annual Meeting of the Transportation Research Board (TRB)*, 2025. Lectern Session: Emerging Behaviors of Urban Rail Transit Passengers - A Global Scan.

Nguyen, D. V. A., R. Piplani, A. Gunawan. GRASP reinforced by Evolutionary Path-relinking for two-echelon covering tour vehicle routing problem. *33rd European Conference on Operational Research (EURO)*, 2024.

Thesis

Nguyen, D. V. A., (2022). Two-echelon vehicle routing problem for post-disaster aid. *Nanyang Technological University*.

Funding & Grants

William Demant Foundation, Research Stay Grant for a six-month research stay at the *Technical University of Munich (TUM)*, Germany (DKK 18,500).

Otto Mønsted Foundation, Research Stay Grant for a six-month research stay at the *Technical University of Munich (TUM)*, Germany (DKK 30,000).

Stibo Foundation, Research Stay Grant for a six-month research stay at the *Technical University of Munich (TUM)*, Germany (DKK 30,000).

Otto Mønsted Foundation, Travel Grant for participation in the *Transportation Research Board (TRB) Annual Meeting 2025*, Washington, D.C., USA (DKK 7,500).

Teaching and Mentoring

Teaching Assistant – *42101 Introduction to Operations Research*, Spring 2024, Technical University of Denmark (DTU).

Teaching Assistant – *42578 Advanced Business Analytics*, Spring 2024 & Spring 2025, Technical University of Denmark (DTU).

Teaching Assistant – *42577 Introduction to Business Analytics*, Autumn 2024 & Autumn 2025, Technical University of Denmark (DTU).

Co-supervisor – MSc Thesis: Xiaowei Hou. MSc in Autonomous Systems, Fall 2025. Thesis: *Model-based Reinforcement Learning for Traffic Signal Control*.

Co-supervisor – MSc Thesis: Alma Fazlagic & Kristine Pryds Loft. MSc in Business Analytics, Spring 2025. Thesis: *Forecasting Reconciliation with Origin-Destination Models Using Graph Neural Networks*.

Co-supervisor – MSc Thesis: Damien Fleutry. MSc in Transport and Logistics, Spring 2024. Thesis: *Deep Reinforcement Learning for Traffic Signal Control: Adapting the framework for SimMobility*.

Academic Service

Journal Reviewing

Reviewer for: Neurocomputing (NEUCOM), Artificial Intelligence for Transportation (AIT), Engineering Applications of Artificial Intelligence (EAAI), European Journal of Operational Research (EJOR), Transportation Research Part C: Emerging Technologies (TRC), Transportation Research Part E: Logistics and Transportation Review (TRE).

Conference Reviewing

The Conference on Neural Information Processing Systems (NeurIPS) – 2025, 2026, Transportation Research Board (TRB) Annual Meeting – 2025, The International Conference on Optimization, Modeling, Simulation, and Analytics – 2025, TRC-30: 30th Anniversary of Transportation Research Part C – 2024, Symposium of the European Association for Research in Transportation (hEART) – 2024.

Technical Skills

Programming Languages: Python, MATLAB, Julia

Deep Learning & Graph Learning: PyTorch, PyTorch Geometric, JAX, graph neural networks, GAT/GATv2, LSTM, mixture-of-experts, deep learning

Model-Based RL & Sequential Decision-Making: Reinforcement learning, model-based reinforcement learning, world models, approximate policy iteration, policy-value networks, offline-to-online learning, multi-agent systems

Planning, Control & Optimization: Model predictive control, distributed MPC, receding-horizon control, iterated best response, beam search, combinatorial optimization, mathematical programming, stochastic routing

Optimization Software: Gurobi, CPLEX, OR-Tools

Simulation & Networked Systems: SUMO, TraCI, NetworkX

Research Computing and Software Development: Linux, Git, GitHub, CUDA, multi-GPU training, SLURM, high-performance computing clusters

Professional Memberships

Sep 2025 – Present	IEEE Robotics and Automation Society (IEEE-RAS)
Sep 2025 – Present	IEEE Intelligent Transportation Systems Society (IEEE-ITSS)
Sep 2025 – Present	IEEE Control Systems Society (IEEE-CSS)
Sep 2025 – Present	European Complex Systems Society (ECSS)
Jun 2023 – Present	INFORMS – The Institute for Operations Research and the Management Sciences

References

Associate Professor Filipe Rodrigues

Technical University of Denmark

PhD Supervisor

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Associate Professor Carlos Lima Azevedo

Technical University of Denmark

PhD Co-Supervisor

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Professor Tomer Toledo

Technion - Israel Institute of Technology

Collaborator

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